Hybrid Spatial-Frequency CNN with Attention for Deepfake Detection

Name: Youssef Sufian Zuaiter
Course Name: Deep Learning

Description

Deepfake technology has advanced rapidly in recent years, enabling the creation of highly realistic manipulated videos and images. While these techniques have useful applications in entertainment and media, they also pose serious threats such as misinformation, identity fraud, and security risks. Developing robust methods to automatically detect deepfake content has therefore become an important research challenge in artificial intelligence.

This project develops a hybrid deep learning framework for detecting deepfake images by combining two complementary feature representations: spatial RGB features extracted via a pretrained MobileNetV2 backbone enhanced with a channel-wise attention mechanism, and frequency-domain features derived from a 2D Fast Fourier Transform (FFT) branch. The core insight is that GAN-based deepfake generation pipelines introduce periodic spectral artifacts that are invisible in RGB space but detectable in the frequency domain, while spatial attention helps the model focus on blending boundaries and texture inconsistencies in facial regions.

Dataset

The FaceForensics++ C23 dataset is used, obtained from Kaggle (gradientvoyager/faceforensics-c23-extracted-faces-100k). It contains approximately 182,000 pre-extracted face images across seven classes: Real, Deepfakes, Face2Face, FaceShifter, FaceSwap, NeuralTextures, and DeepFakeDetection. Labels are collapsed into binary form (Real = 0, Fake = 1) for classification. The dataset is split into training (128,477), validation (27,493), and test (26,276) sets. Since fake images outnumber real images by approximately 6:1, balanced undersampling is applied during training to prevent class-imbalance bias.

Methodology

The project follows a structured ablation study across three models of increasing complexity:

> **Stage 1 — Baseline MobileNetV2:** A pretrained MobileNetV2 model with its final classifier replaced by a 2-neuron binary head. All backbone weights are fine-tuned to adapt ImageNet features to deepfake artifacts. This serves as the reference model for the ablation study.

Stage 2 — Attention-Enhanced CNN: A channel-wise attention module (1x1 Conv + Sigmoid gate) is inserted after the MobileNetV2 feature extractor. The gate re-weights each of the 1,280 feature channels, amplifying channels that encode manipulation artifacts and suppressing irrelevant background features. Grad-CAM visualizations are used to confirm that attention activations concentrate on manipulation-relevant facial regions.

Stage 3 — Hybrid Spatial-Frequency CNN (proposed model): The attention-enhanced spatial branch is combined with a dedicated frequency branch. Each input image is converted to its log-scaled FFT magnitude spectrum, which is processed by a lightweight CNN (3 convolutional blocks with BatchNorm and MaxPooling). The 1,280-dimensional spatial vector and 64-dimensional frequency vector are concatenated and passed through a dense classification head (1,344 -> 256 -> 64 -> 2) with Dropout regularization.

All models are trained with the Adam optimizer (lr = 1e-4), CrossEntropyLoss with label smoothing (0.1), and a ReduceLROnPlateau scheduler that halves the learning rate when validation accuracy stagnates. Training transforms include RandomHorizontalFlip, RandomRotation, and ColorJitter augmentation alongside ImageNet normalization. The best model checkpoint is saved based on validation accuracy.

Evaluation

Models are evaluated on the held-out test set using accuracy, precision, recall, F1-score, confusion matrix, and AUC-ROC. Training and validation learning curves are plotted to assess convergence and overfitting. Grad-CAM visualizations are generated to interpret spatial attention patterns. The three-model ablation study quantifies the isolated contribution of each architectural component.

Expected Results

Model	Val Accuracy	Test Accuracy	vs Baseline
Baseline MobileNetV2	87.52%	88.59%	—
Attention CNN	90.78%	91.03%	+2.44%
Hybrid Spatial-Frequency CNN	**92.83%**	**93.34%**	**+4.75%**

Contributions

The main contributions of this project are: (1) a novel hybrid architecture that combines spatial RGB features with frequency-domain FFT features for deepfake detection; (2) a systematic three-model ablation study demonstrating the isolated impact of attention and frequency components; (3) comprehensive evaluation including AUC-ROC analysis and Grad-CAM interpretability; and (4) a practical lightweight implementation achieving 93.34% test accuracy on FaceForensics++ C23 without requiring transformer-scale compute resources.

Limitations & Future Work

The model is trained and evaluated within the FaceForensics++ distribution; cross-dataset generalization to DFDC or Celeb-DF remains untested. The Real class precision (0.74) reflects residual imbalance in the test set and could be improved with focal loss or hard-negative mining. Future work includes replacing channel attention with CBAM, testing EfficientNet or ViT backbones, and evaluating cross-dataset robustness.